

Heric Corrêa da Silva

3rd year PhD Student, IMPA.

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Summary

My research primarily focuses on **Dynamical Systems, Ergodic Theory, and Low-dimensional Geometry and Topology**, with a particular emphasis on low-dimensional dynamics and rotation theory on surfaces. My current work explores periodic-point-free homeomorphisms on the torus beyond the isotopic class of the identity, specifically investigating the existence of irrational circle factors for conservative homeomorphisms homotopic to Dehn twists and an effective forcing theory in this setting. Additionally, I have a solid background in Aubry-Mather and weak-KAM theory within the context of Lagrangian and Hamiltonian dynamics.

Education

Since 2023	Doutorado em Matemática (PhD) Instituto de Matemática Pura e Aplicada (IMPA), Rio de Janeiro, Brazil. Advisor: Alejandro Kocsard
2021–2023	Mestrado em Matemática (MSc) Universidade Federal do Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil. Thesis title: <i>Uniform hyperbolicity and fibered rotation numbers for one parameter linear cocycles</i> Advisor: Katrin Gelfert
2018–2021	Bacharelado em Matemática (BSc) Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brazil.

Preprints

[1] Polynomial entropy and Rotation Sets, *Permanent preprint* [\[Link\]](#).

Undergraduating research articles

- [1] Sobre a existência de medidas invariantes que minimizam a ação de Mather, *Rev.Mat. de Ouro Preto*, vol.2:101-116, 2021. [\[Link\]](#). Title in english: “On the existence of invariant measures that minimizes the Mather’s action”.
- [2] Equidistribuição e lei de Benford em progressões geométricas, *Rev.Mat de Ouro Preto*, vol.3:76-86, 2022. [\[Link\]](#) Title in english: “Equidistribution and Benford’s law in geometric progression”.

Teaching Experience

2026	Substitute Professor Calculus for Economics	UFRJ
2025	Teaching Assistant Smooth Ergodic Theory	IMPA
2023	Teaching Assistant Non-euclidean geometry	UFRJ
2021	Teaching Assistant Calculus of several variables	UFMG

Scholarships

Since 2023	CNPq scholarship for PhD
2021-2022	CAPES scholarship for MSc
2015-2017	FAPEMIG scholarship for undergraduating research projects

Academic Events Attended

2025	Beyond Uniform Hyperbolicity I and II	Trieste, Italy.
2025	8th Brazilian School in Dynamical Systems	Rio de Janeiro, Brazil.
2023	1st LESET Lyapunov Exponent and Smooth Ergodic Theory Encounter	Paraty, Brazil.
2023	7th Brazilian School in Dynamical Systems	Fortaleza, Brazil.
2022	Escuela de foliacion y flujo Anosov de dimension 3	Montevideo, Uruguay.
2022	Int. Conference on Dynamical Systems 60th birthday of Marcelo Viana	Rio de Janeiro, Brazil.
2022	A Living Singularity Celebrating Maria José Pacifico's 70th birthday	Virtual.
2021	6th Brazilian School in Dynamical Systems	Virtual
2019	5th Brazilian School in Dynamical Systems	Belo Horizonte, Brazil.
2018	WEAK KAM in Rio 2018, ICM Satellite Conference	Rio de Janeiro, Brazil.

Outreach and Volunteer Services

II Festival Nacional da Matemática (National Mathematics Festival)

Rio de Janeiro, Brazil

I was an exhibitor of the mathematical artworks of the Imaginary organisation (www.imaginary.org) joint with IMPA.

Oct. 2022

NGO Sou-Crânio Preparatório

Belo Horizonte, Brazil

I was a teacher in a non-governmental organization which aims to prepare students in social vulnerability conditions for university admission exams in Brazil.

2017-2020

References

- Prof. Katrin Gelfert
Professor at Universidade Federal do Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil
✉ gelfert@im.ufrj.br
- Prof. Alejandro Kocsard
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